Gearbox Housing Assembly - Cover and Flange Bolting

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Document type	Work Instruction	**Status**	Superseded
Effective date	2025-03-10	**Owner**	Production Engineering
Classification	Internal		

1 Purpose

This work instruction describes how the gearbox housing cover is sealed and bolted to the housing on assembly line L2. It applies to the e-drive variants ED-150 and ED-210.

2 Safety

Wear cut-resistant gloves when handling the housing cover. Wear nitrile gloves when working with the sealant. The EC screwdriver may only be used with the reaction arm fitted.

3 Tightening torques

Joint	Thread	Torque	Tolerance
Housing cover bolts	M8	25 Nm	+/- 2.5 Nm
Flange bolts to motor housing	M10	40 Nm	+/- 4 Nm
Speed sensor bracket	M6	9 Nm	+/- 0.9 Nm
Oil drain plug	M14x1.5	30 Nm	+/- 3 Nm

4 Procedure

1 Check the EC screwdriver with the torque analyser at the start of every shift.

2 Clean the flange surfaces with a lint-free cloth and cleaning agent C-1.

3 Apply a continuous sealant bead of 2.5 mm +/- 0.5 mm.

4 Fit the cover within 15 minutes of applying the sealant.

5 Tighten the bolts in a cross pattern in two stages: 50 % and 100 % of the final torque.

6 Let the sealant cure for at least 20 minutes before the leak test.

5 Revision history

Rev	Date	Change
A	2024-09-02	First issue
B	2025-03-10	Drain plug added

SUPERSEDED